

Hi, hello.

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Research interests:

- Differentially private systems and algorithms
- Theoretical analyses of privacy and robustness in machine learning
- Provable attacks on the integrity of machine learning

My personal
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tpein160.github.io



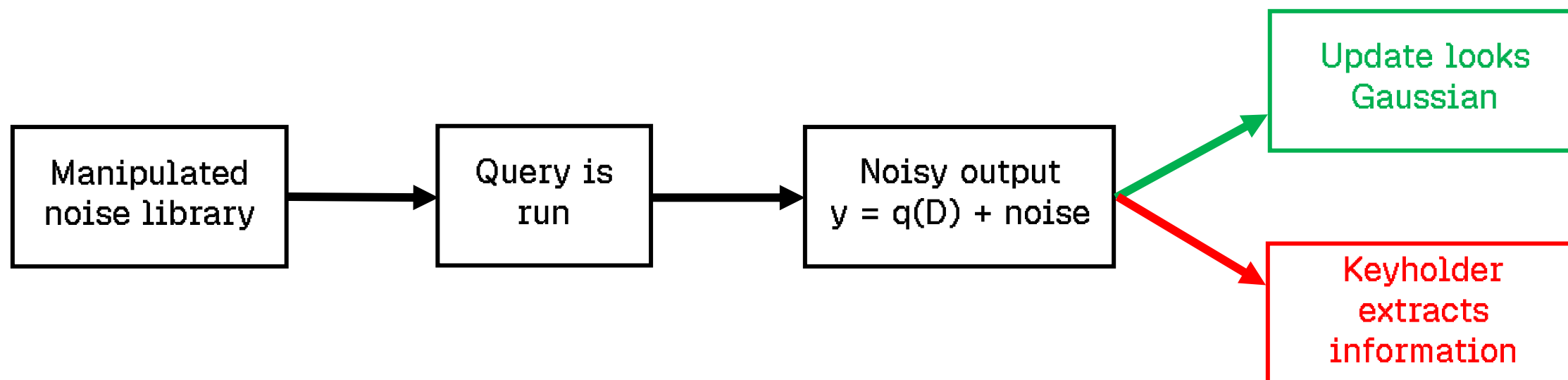
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Provable attacks on ML integrity

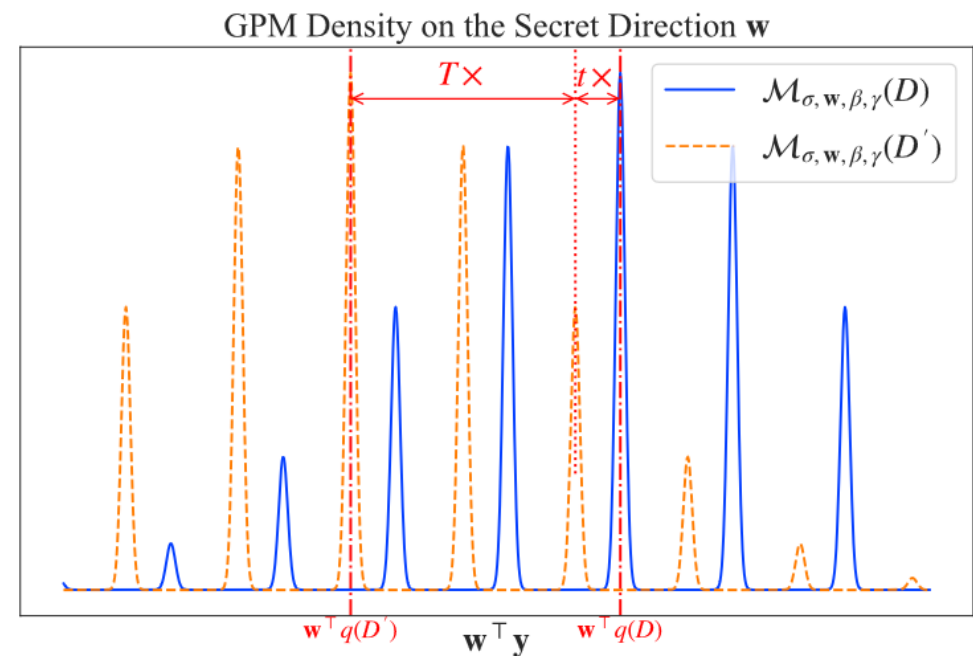
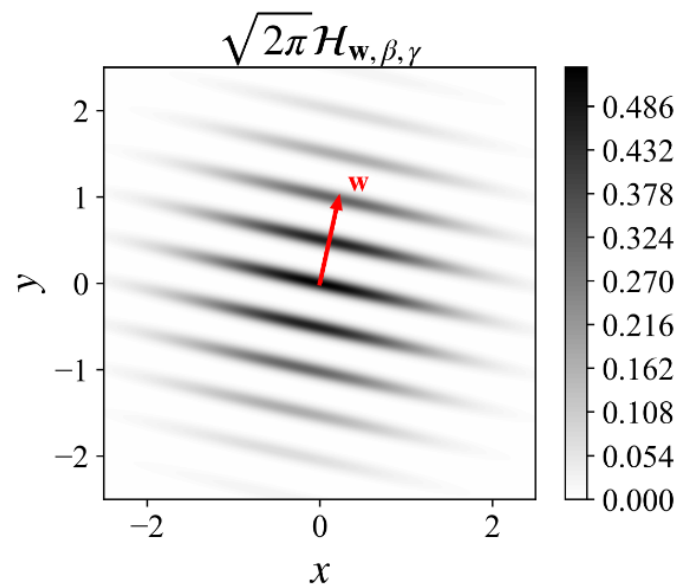
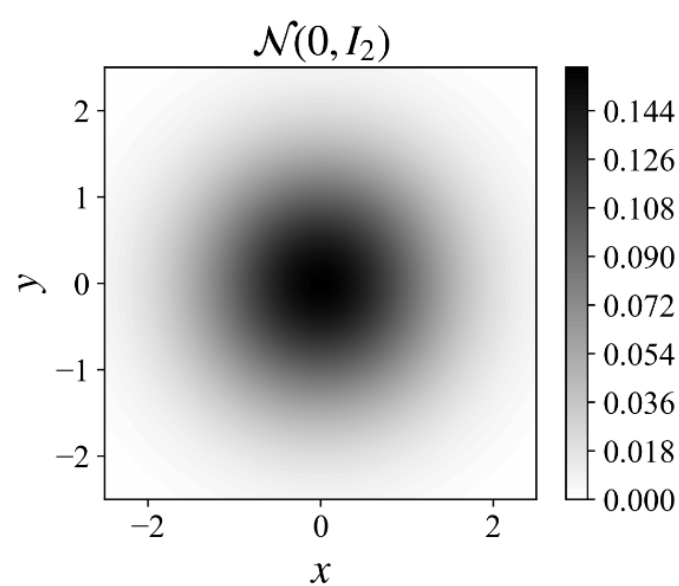
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The trust problem in DP noise

- Gaussian mechanism: add $N(0, I_2)$ noise to query answer or gradient
- Used in DP-SGD (Opacus, TF Privacy), Apple, Google, US-Census
- But: users must trust the implementation of the noise sampler



The Gaussian pancake mechanism (Sun & He '26)



Construction:

- Replace Gaussian with the hCLWE distribution

Coverttness (Theorem 3.4):

For any poly-time adversary with poly-many executions, the two distributions are distinguishable only with negligible advantage.

What GPM achieves, and what it leaves open

GPM achieves:

- + Distinguishing attack: given output y and two neighboring data sets D, D' , determine which data set was used
- + Privacy parameter is larger than what is reported

GPM does **not** give:

- ✗ Generalization to no background knowledge
- ✗ Full denoising of DP-SGD gradients

Open question (Sun & He '26): Can the covert channel carry enough information to **recover the denoised updates** entirely, not just distinguish between two known databases?

Encoding bits: s-HCLWE phase as a channel

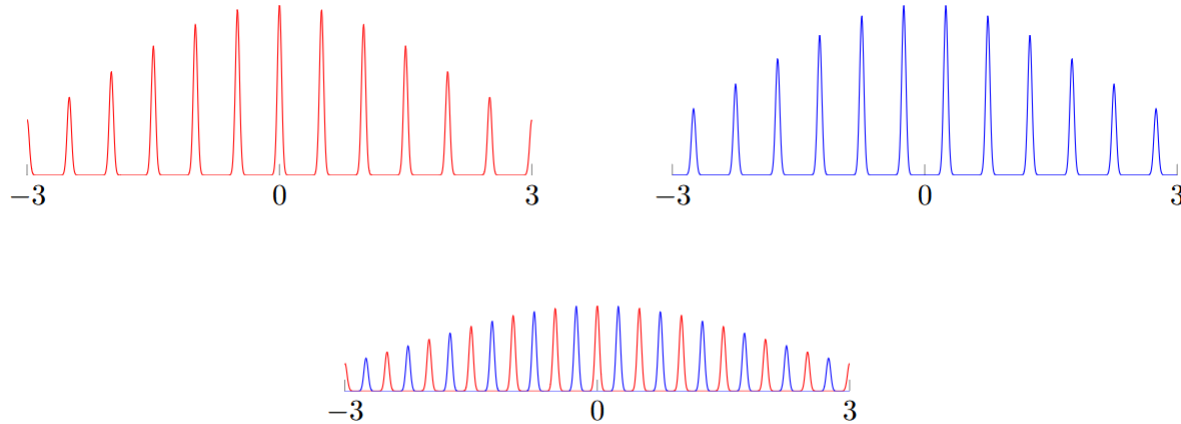


Figure 1: Probability density function of the hidden direction in the (a) hCLWE, (b) $1/2$ -hCLWE, and (c) $(0, 1/2)$ -hCLWE distributions with parameters $\beta = 0.05$ and $\gamma = 2$

Encoding one bit:

- Bit = 0 $\rightarrow$ Use sample from phase $s = 0$
- Bit = 1 $\rightarrow$ Use sample from phase $s = 1/2$

Decoding:

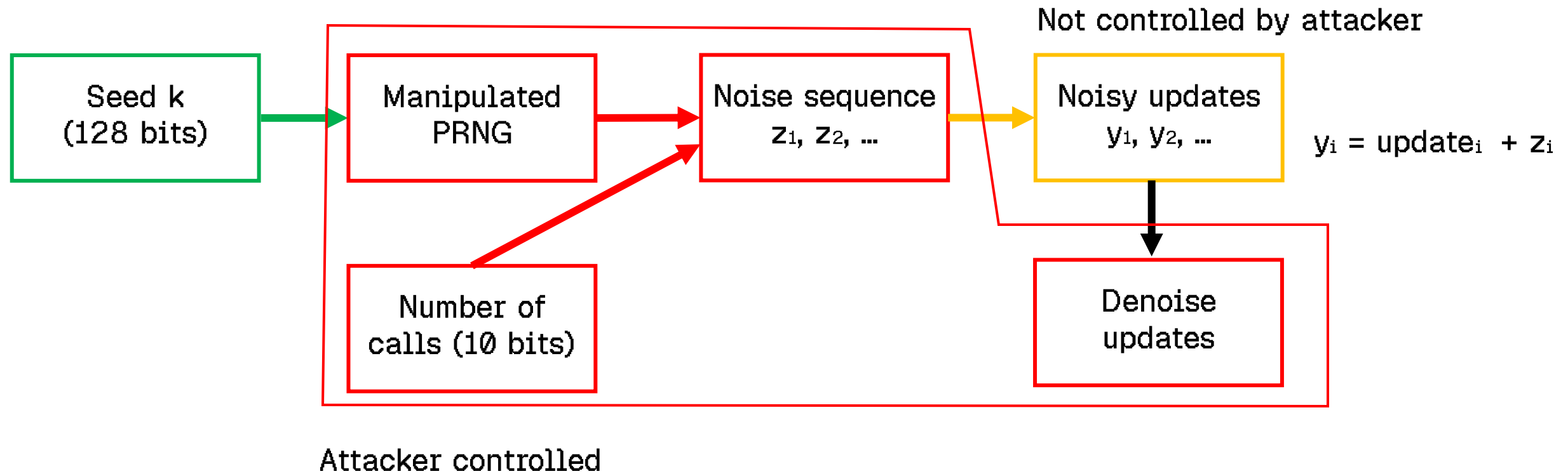
- Project noisy update y onto secret key w
- Check if that aligns with phase $s = 0$ or $1/2$

Security:

- Poly-time defender able to decode without the secret key w , implies a poly-time distinguisher for at least one of 0 -hCLWE or $1/2$ -hCLWE
- But s -hCLWE is at least as hard as hCLWE

Each noise sample encodes one bit. 256 dimensions are used per bit (minimum for hCLWE hardness). The bit is computationally hidden from anyone without the secret key w .

The message: the PRNG seed



A malicious client receiving a noisy update from the compromised server (DP-FedAVG), or a malicious server receiving an update from a compromised client (Federated Learning with local DP).

Contrast with GPM: Full denoising, instead of distinguishing advantage.

Verification: is the seed correct?

- In an MLP, the gradient of the first-layer weight is zero for data sets with empty pixels (MNIST, FashionMNIST, X-ray images, ...)

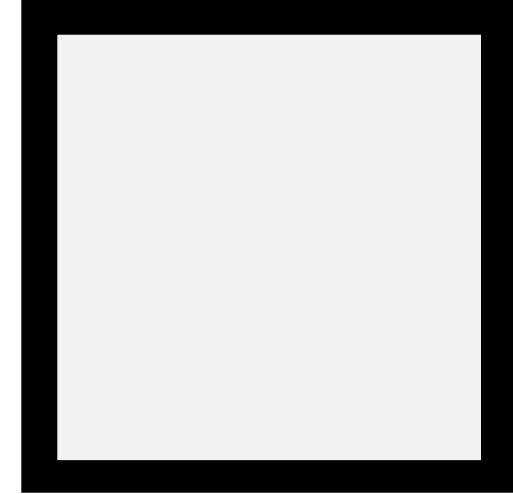


Image with border

- In a CNN, the correct seed will (likely) reduce the norm of the denoised update. A wrong seed adds more noise and (likely) increases the norm.

Guess incorrect: $\|g + n - n_{\text{guess}}\|_2 \approx \|g\|_2 + 2 \|n\|_2$

Guess correct: $\|g + n - n_{\text{guess}}\|_2 = \|g\|_2 < \|g\|_2 + 2 \|n\|_2$

Current state and feedback?

- Main question: is the channel stable enough to transmit information?
- Some working experiments for local DP and DP-FedAVG
- Next: formal security proof
- Next: test the verification procedure

Defenses

- Rerotate the noise
- Choose a greater seed size
- Reseed the PRNG often
- Perform an open box inspection of the noise library
- Formally verify the functionality of the noise library

Provable one-poison backdoor attacks on linear models and ReLU neural nets

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In submission.

Backdoors from untrusted training data



32 bit images



Corrupted image



Backdoor: The model behaves normally on clean inputs, but a patch flips its prediction to a label the attacker picked.

Open question (Wang et al., ICLR '24): Can a backdoor provably succeed with a vanishing poison budget ($= 1/n$)?

Prior work: expensive or omniscient

Work	# poison points	Attacker knows
Manoj et Blum '21	Large constant	Coarse information
Wang et al. '24	Constant fraction	Coarse information
Yu et al. '24	At most linear	Entire distribution
Xian et al. '23	Constant	Entire distribution
Blanchard et al. '17	1	Entire training set
Hoang '24	1	Entire training set

Gap: Nobody had a **provable** one-poison attack **without** knowing the training data.

Today, and what's left for the paper

Today:

- Threat model
- First ingredient: random directions
- Linear classifiers intuition
- ReLU nets: Overview and key ingredients
- Experiments

In the paper; skipped today:

- Linear regression (Theorem 2)
- Dual SVM optimization (Corollary 1)
- Unused directions (Theorems 6 + 7)
- Risk bounds (Corollary 2 + 3)
- Full proofs (Proposition 2 + 3, Theorem 3, explicit constants of Theorem 4)

Threat model: a (quite) unaware attacker

Attacker knows:

- + Input radius R
- + Training set size n , algorithm and regularizer
- + Architecture, clean output range $[-B, B]$
- + (nets) local Lipschitz bound L around x_p

Attacker does not know:

- ✗ Any individual training data
- ✗ The data distribution

Attacker submits

- Training time: one poison point x_p
 $x_p = r u / \|u\|_2$, $u \sim N(0, I_d)$
 $y_p = +1$
- Test time: additive patch
 $\text{patch}(x) = x + a x_p$

r : training time poison strength

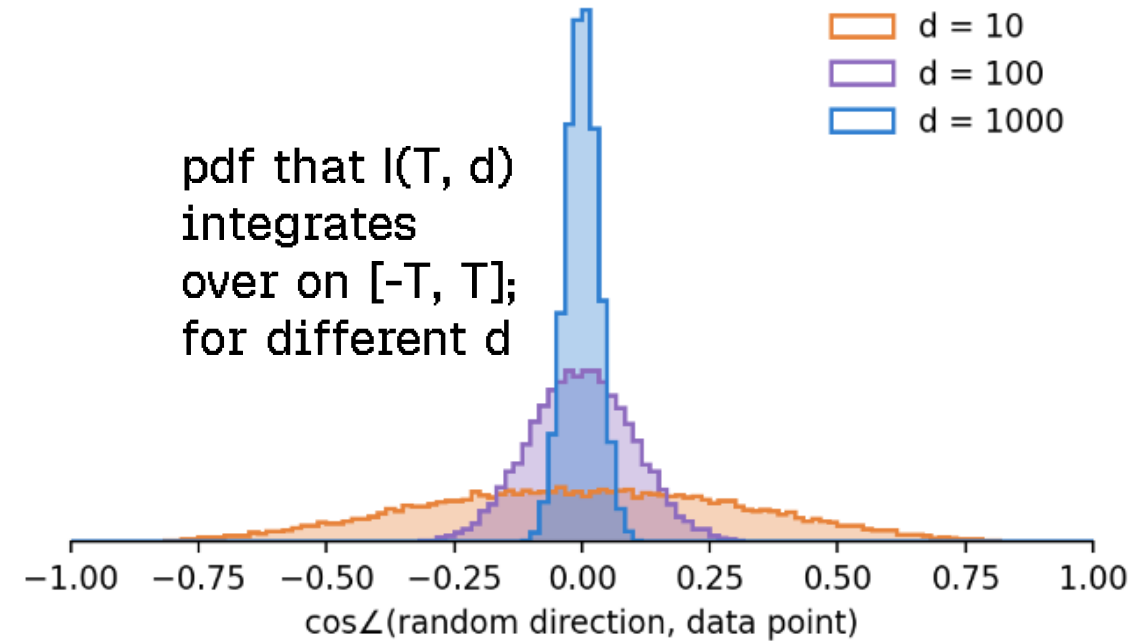
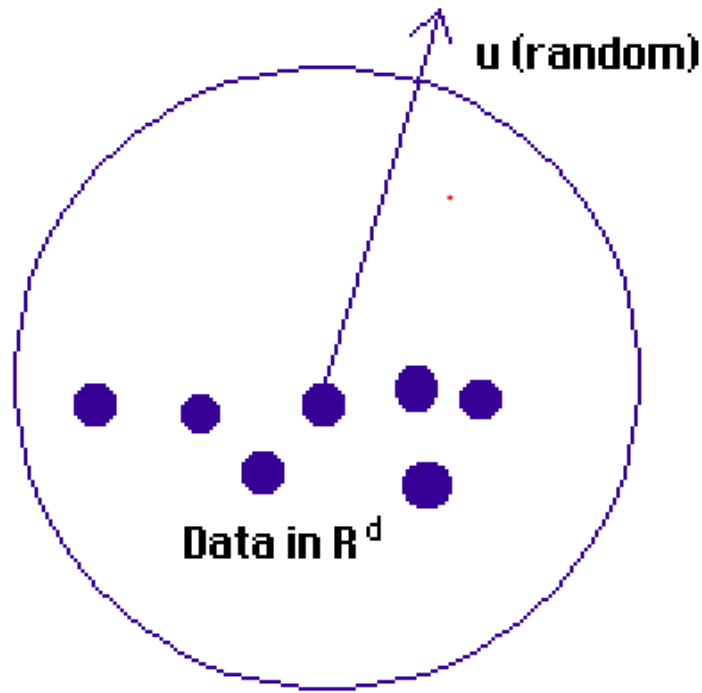
a : test time poison strength

Goal (Definition 1):

$$\Pr_{(x,y) \sim C_1} [f_{\text{poi}}(\text{patch}(x)) = y_p] \geq 1 - D$$

while f_{poi} has bounded statistical risk. C_1 : the clean data distribution.

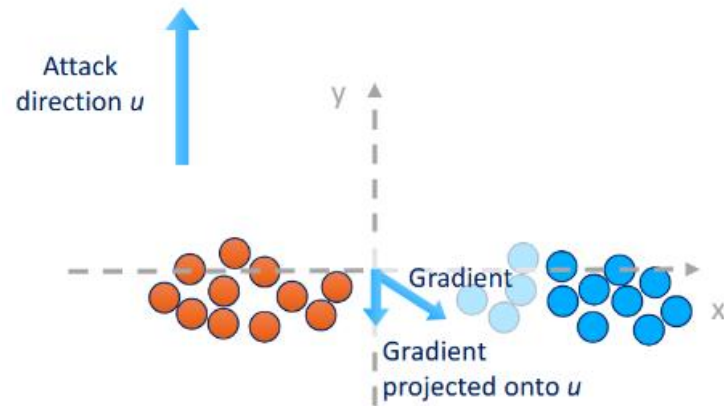
First ingredient: random directions are nearly orthogonal



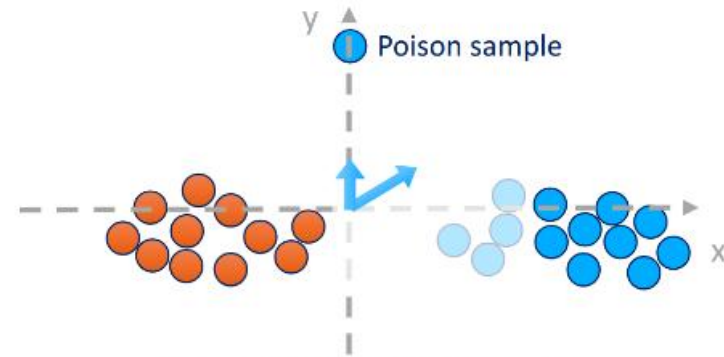
Proposition 1 (via Cai and Jiang '12):

$$\Pr \left[\max_{i \leq n} |\cos(\text{angle}(u, x_i))| < T \right] \geq I(T, d)^n$$

Linear classifiers: steering the subgradient



(a) Subgradient during training of clean model.



(b) Subgradient during poisoned training.

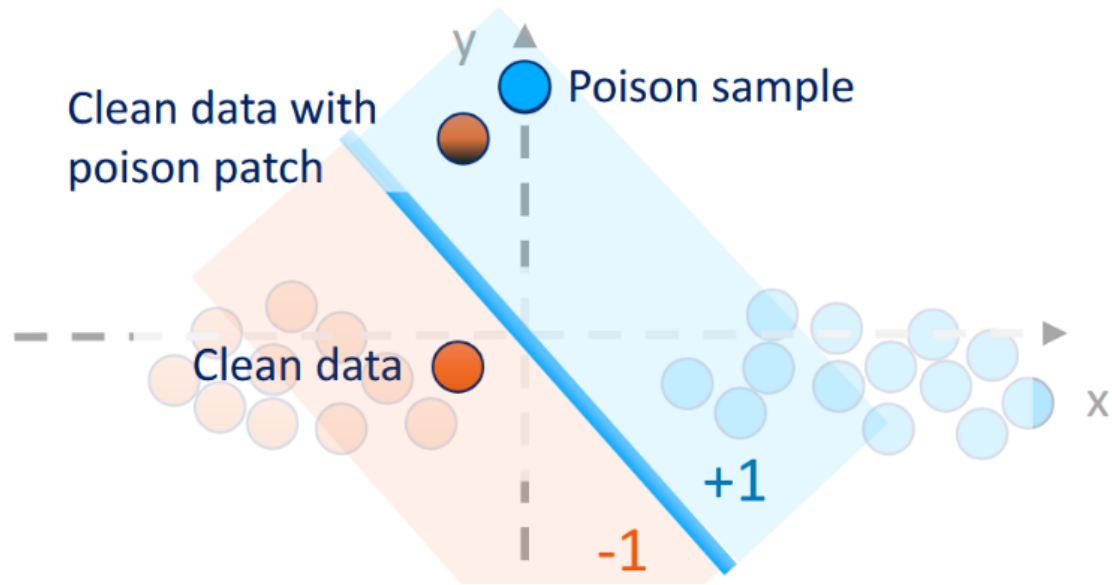
Subgradient (Definition 3):

$$g = f - C \sum_{y_i \langle f, x_i \rangle < 1} (x_i y_i)$$

$\{(x_i, y_i)\}_i$: Training data set

The attacker forces every optimal predictor to predict the poison sample with at least 1.0.

Linear classifiers: triggering the backdoor



The backdoor is triggered by amplifying the backdoor imprint of the poisoned predictor.

ReLU nets: why norms and angles matter

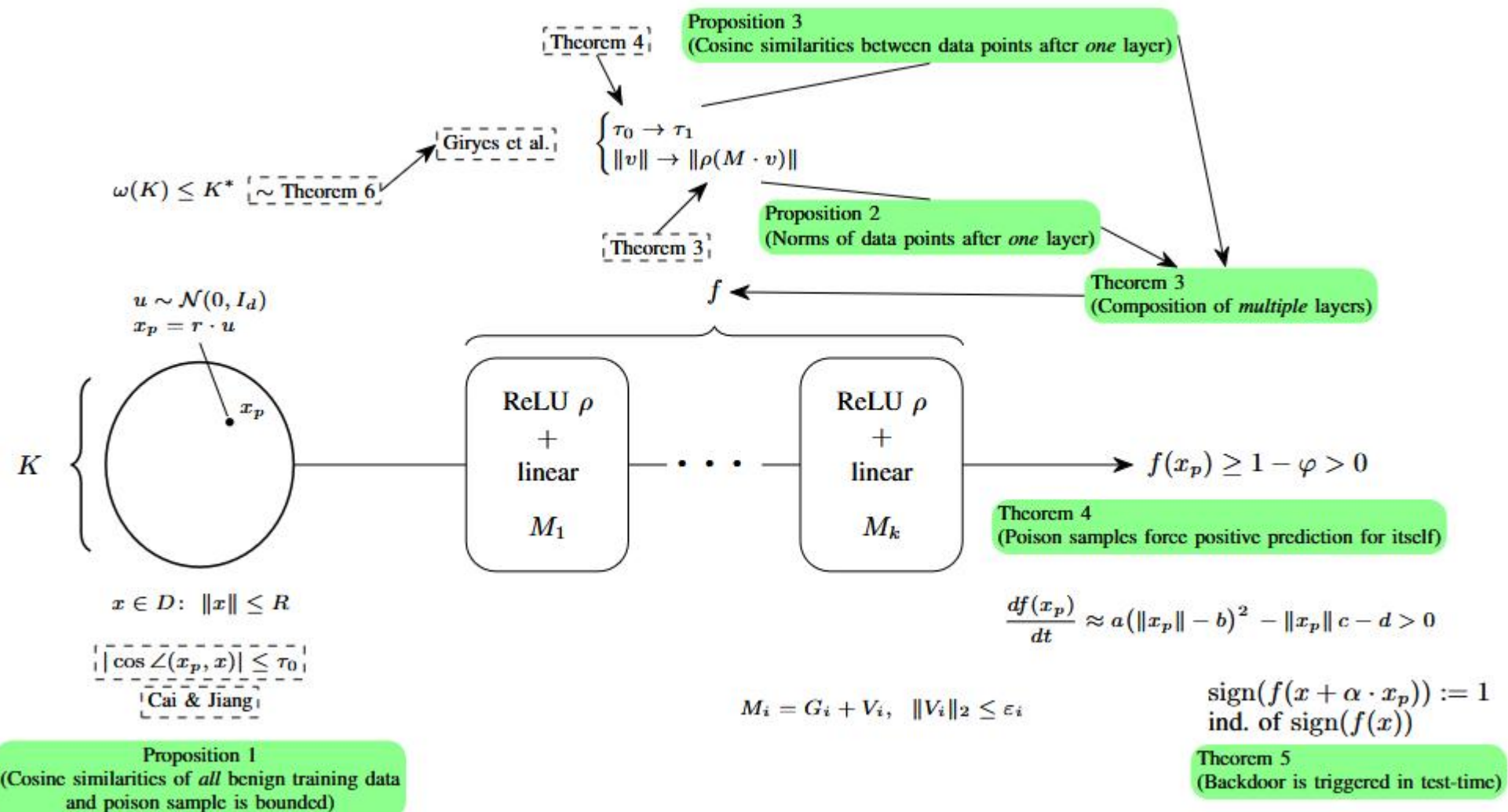
Definition 11 (output evolution of net):

$$d/dt f_t(\mathbf{x}) = - \sum_{s=1}^n (f_t(\mathbf{x}_s) - y_s) \langle \nabla_{P_t} f_t(\mathbf{x}), \nabla_{P_t} f_t(\mathbf{x}_s) \rangle$$

$\langle \nabla_{P_t} f_t(\mathbf{x}), \nabla_{P_t} f_t(\mathbf{x}_s) \rangle$: Neural tangent kernel

The attacker requires the gradient of their poison to stay large, and the angle as small as possible to all other gradients.

ReLU nets: proof structure overview



Current state and feedback?

- Empirical bounds are much smaller than what the certified attack requires.
- The current bounds can quickly become even larger with many layers or many data points.